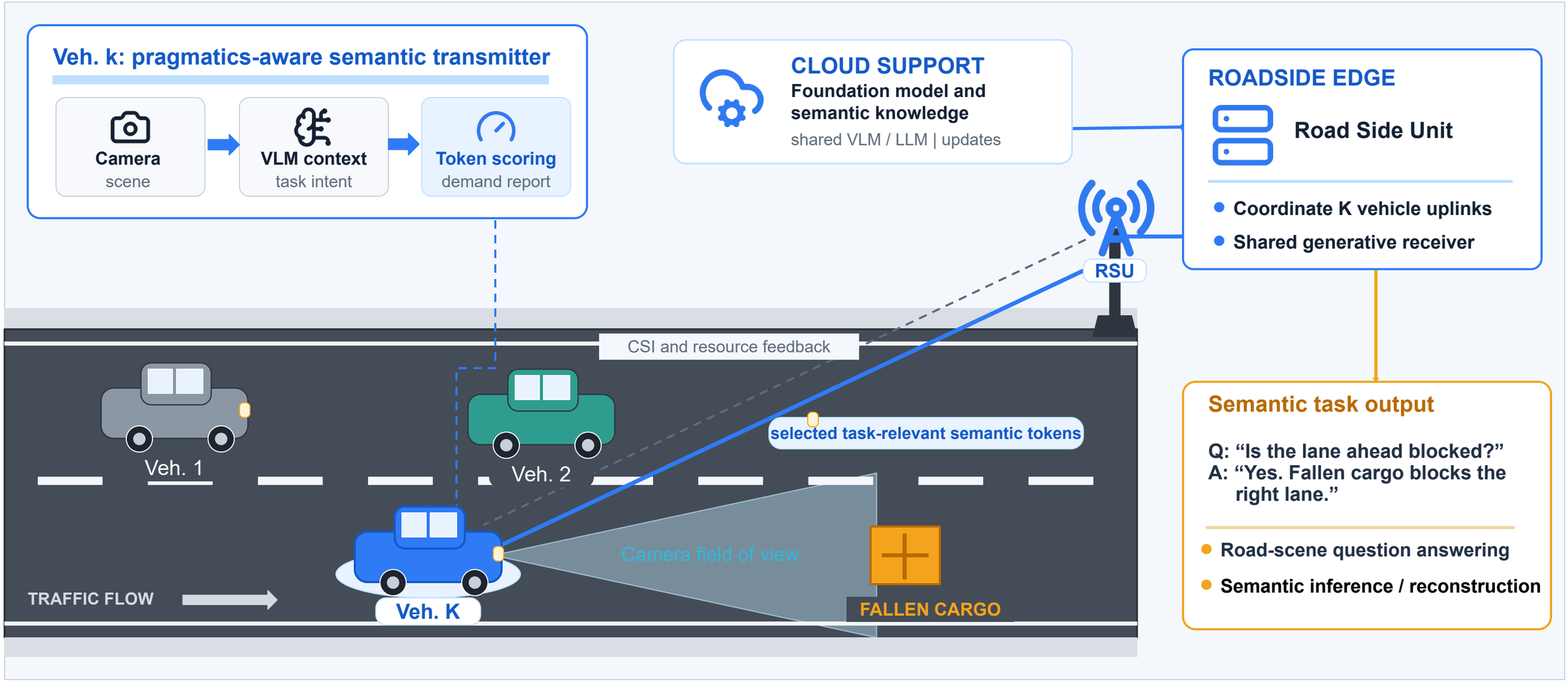

Veh. k: pragmatics-aware semantic transmitter
Camera
scene
VLM context
task intent
Token scoring
demand report
CLOUD SUPPORT
Foundation model and semantic knowledge
shared VLM / LLM | updates
ROADSIDE EDGE
Road Side Unit
Coordinate K vehicle uplinks
Shared generative receiver
RSU
CSI and resource feedback
selected task-relevant semantic tokens
Veh. 1
Veh. 2
Veh. K
Camera field of view
FALLEN CARGO
TRAFFIC FLOW
Semantic task output
Q: “Is the lane ahead blocked?”
A: “Yes. Fallen cargo blocks the right lane.”
Road-scene question answering
Semantic inference / reconstruction